

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445UB0-1



TUESDAY, 13 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life

HIGHER TIER

1 hour 30 minutes

Section A

Section B

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	8	
3.	6	
4.	7	
5.	10	
6.	9	
7.	8	
8.	16	
Total	75	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

- Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
- You may use a pencil for graphs and diagrams only.
- Write your name, centre number and candidate number in the spaces at the top of this page.
- Answer **all** questions.
- Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets at the end of each question or part-question.
- Question **5(a)** is a quality of extended response (QER) question where your writing skills will be assessed.
- You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
- You will need to refer to the separate Resource Folder to answer questions **1, 2 and 3**.
- A Periodic Table is printed on page 20.



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Section A

Answer **all** questions.

Refer to the separate Resource Folder to answer the following questions.

1. (a) Use the information under **Figure 1** to answer the following question.
The wave speed of ultrasound in body tissue is 1540 m/s.

Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of the ultrasound used in a 2 MHz scan.

[3]

wavelength = m

- (b) Use the information in **Figure 4** to answer the following questions.

- (i) Select the most suitable radioisotope of **iodine** from the table in **Figure 4** that would be used as a tracer. Explain your choice.

[2]

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- (ii) The photograph shows a patient undergoing external beam radiation therapy for a brain tumour.



Select the most suitable radioisotope from the table in **Figure 4** that would be used for this treatment. Explain your choice.

[3]

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- (c) Refer to **Figures 4 and 6** in the Resource Folder to answer the following question.

State which of the radioisotopes **R1, R2, R3, R4, R5** or **R6**, is lutetium-177. **Explain** your reasoning.

[3]

Radioisotope is lutetium-177.

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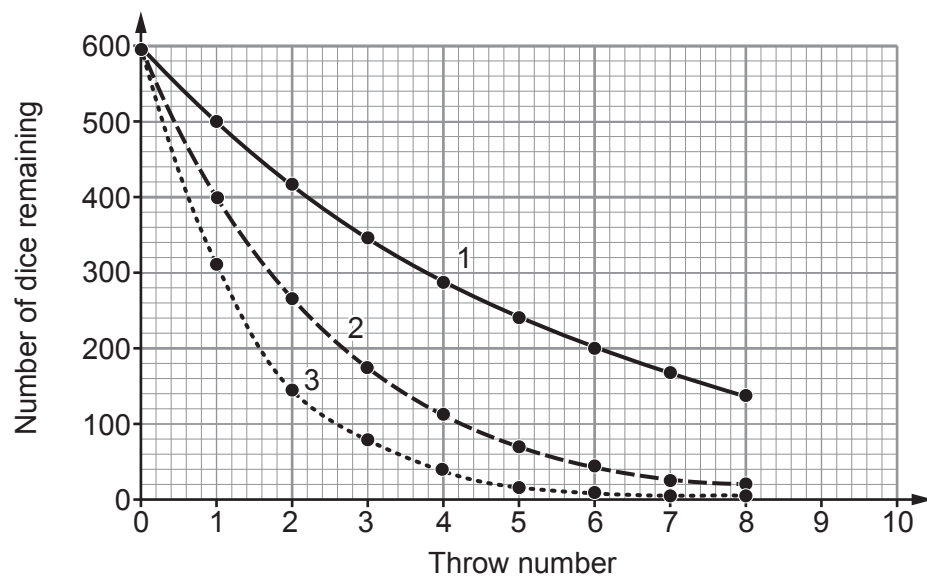
2. The following questions are based on the information on pages 6 and 7 of the Resource Folder about using dice to model half-life.

- (a) Each group's results were added together to give class results. Give **one** reason why this makes the data more repeatable. [1]

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- (b) The results in **Figures 7 and 8** in the Resource Folder are used to plot graphs on the grid below.



- (i) Add lines to the graph to determine the half-life represented by line 3. [2]

half-life = throws



- (ii) Use the information in the graph to **tick (✓)** the boxes next to the **three** correct statements. [3]

Line 1 represents the data from Model 2.

☐

Line 2 shows decay of the quickest rate.

☐

Line 2 shows the equivalent of three half-lives after 5 throws.

☐

Line 1 represents the longest half-life.

☐

Line 3 would have been produced if 600 coins had been used instead of dice and the number of 'heads' counted after each throw.

☐

Line 3 has an activity that is $\frac{1}{8}$ of the original after 8 throws.

☐

- (c) The teacher suggests to the class that they add 5 red wooden cubes to the 50 dice they start with. The red cubes would not be removed after each throw but would be counted every time to improve the modelling of radioactive decay.

- (i) State what the red cubes represent. [1]

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- (ii) State how the presence of the red cubes would affect line 1 on the graph. [1]

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3. X-rays are commonly used to diagnose illness.

(a) Refer to **Figure 2** to answer the following.

It is stated that X-rays have shorter wavelengths than ultraviolet rays but longer wavelengths than gamma rays. Explain whether you agree using data to support your answer. [3]

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(b) Refer to **Figure 3** to answer the following.

A patient has a series of X-rays taken following an accident. These were: one chest; one abdomen; and one skull X-ray. Calculate the percentage of the yearly background radiation the patient received from this combination of X-rays. [3]
(1 year = 365 days)

percentage =



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Section B

Answer **all** questions.

4. Lichen can be used as an indicator species for air pollutants such as sulfur dioxide. Students performed an air pollution survey by recording the distribution of lichen around a city. Their results are shown in the table.

Distance from city centre (km)	Abundance of lichen (units)	Mean concentration of sulfur dioxide (units)
0.0	0	250
1.3	3	198
2.5	6	160
3.6	10	140
5.4	14	122

- (a) Use the information in the table to plot a graph showing the change in mean concentration of sulfur dioxide with the distance from the city centre on the grid opposite and draw a suitable line. [3]



Mean concentration of sulfur dioxide (units)

100
0

Distance from city centre (km)

3445UB01
09



- (b) Describe the relationship between distance from the city centre and mean concentration of sulfur dioxide. [2]

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- (c) Explain the effect of sulfur dioxide on the abundance of lichen. [2]

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Examiner
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5. (a) Explain why, for thousands of years, the percentage of carbon dioxide in the Earth's atmosphere remained approximately constant **and** how this stabilised conditions for life on Earth. [6 QER]

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- (b) During the past 200 years, evidence shows that the percentage of carbon dioxide in the atmosphere has been steadily rising.

One factor that contributes to this increase is how people travel to work.

The table shown below is taken from a climate change website.

It gives information about the mass of carbon dioxide produced **each journey per person** travelling to work by different methods each day.

Distance travelled (km)	10	20	25	50
carbon dioxide produced travelling by car (kg)	0.60	1.20	1.50	3.00
carbon dioxide produced travelling by bus (kg)	0.10	0.20	0.25	0.50
carbon dioxide produced travelling by train (kg)	0.18	0.36	0.45	0.90

Use this information to calculate the mass of carbon dioxide saved during a 5-day working week, if **250** people travel **30 km each way** to work and back home by train instead of every person using their own car.

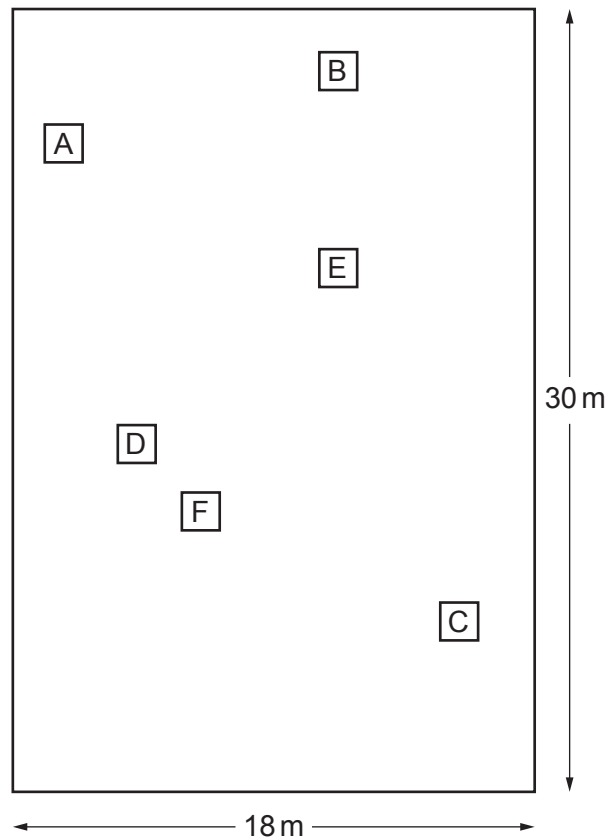
[4]

mass of carbon dioxide saved =kg

10



6. (a) Some students investigated the number of dandelion plants on a lawn. The diagram shows the lawn and the location of 6 quadrats (**A to F**) of area 1 m^2 which the students had placed at random on the lawn.



The students counted the number of dandelions in each quadrat and recorded their results in the table below.

Quadrat	Number of dandelions
A	6
B	3
C	2
D	7
E	4
F	5



- (i) Use the information to estimate the total number of dandelions on the lawn. [3]

estimated number of dandelions =

- (ii) The actual number of dandelions on the lawn is 1950.
Use the formula below to calculate the percentage error of your estimate above. [2]

$$\text{percentage error} = \frac{\text{estimated number of dandelions} - \text{actual number of dandelions}}{\text{actual number of dandelions}} \times 100$$

percentage error =

- (b) A species of dandelion has become resistant to one type of weedkiller. Explain, in terms of natural selection, how this has occurred. [4]

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7. Glucose homeostasis is an important biological process.

(a) Describe the role of insulin in controlling high blood glucose levels.

[4]

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(b) Describe the difference between type 1 and type 2 diabetes.

[2]

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(c) Describe the chemical test for glucose and state the colour change for a positive result.

[2]

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8. Cystic fibrosis is an example of an inherited disease caused by a recessive allele. A carrier of the disease has a genotype Ff.

- (a) Compare the chances of a child being born with cystic fibrosis if both parents are carriers, with a child whose father suffers from cystic fibrosis and a mother who is a carrier. [4]

- (b) Cystic fibrosis causes a build-up of thick mucus in the lungs, which leads to poor gas exchange. Mucus in the lungs also allows increased bacterial growth that is responsible for infections.

Explain how the body defends itself against bacterial infection in the lungs. [3]

- (c) Sufferers of cystic fibrosis benefit from being physically active to slow the rate of decline in lung function.

Explain the short-term and long-term effects of exercise on lung function. [3]

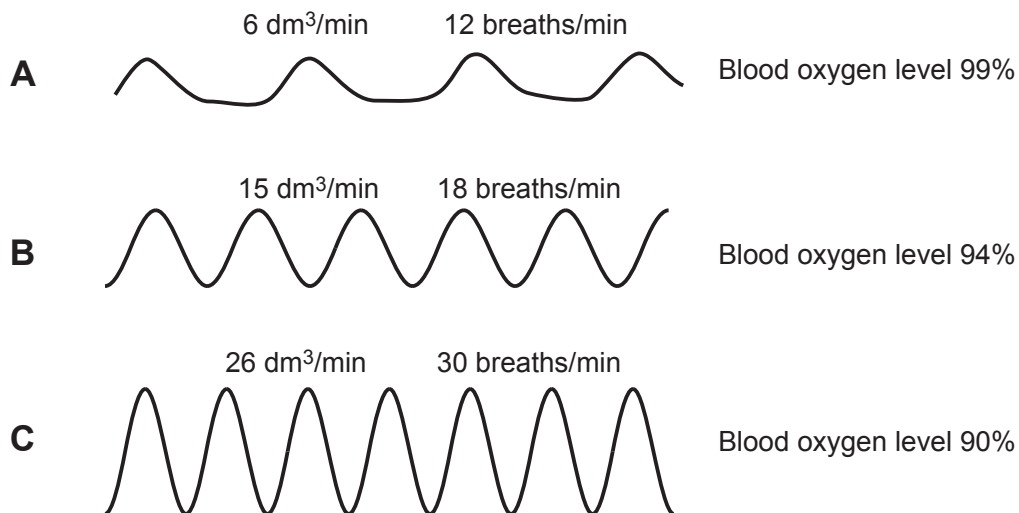


- (d) The diagram below gives information about the breathing of three different individuals, all of whom are resting.

A is a person who is not affected by cystic fibrosis.

B is a person with cystic fibrosis.

C is a person with cystic fibrosis and a lung infection.



The numbers show the volume of air breathed in per minute in dm³ per min and also the number of breaths per minute.

- (i) Use the equation:

$$\text{volume of air per breath} = \frac{\text{volume of air breathed in per minute}}{\text{number of breaths per minute}}$$

to compare the volume of air breathed in per breath for each of the breathing patterns.

[4]

A B C

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(ii) Compare the number of breaths per minute with blood oxygen levels in the three individuals. [2]

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16

END OF PAPER





THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																		
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10	4 He Helium 2
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18	
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36	
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54	
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86	
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																

Key

1

H

Hydrogen

1

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number